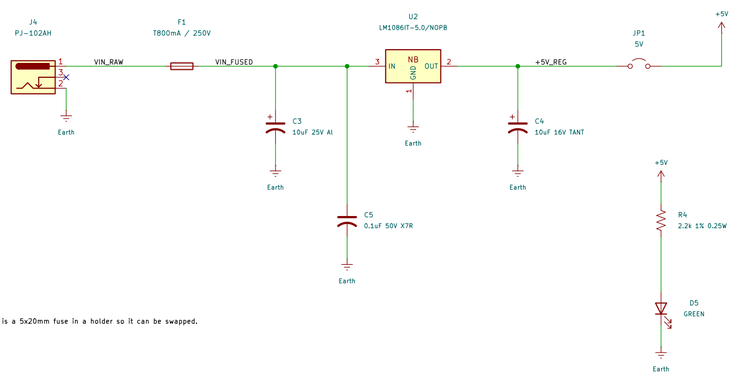
Fill-A-Bot Main Control Board

Power Input and 5V Regulator

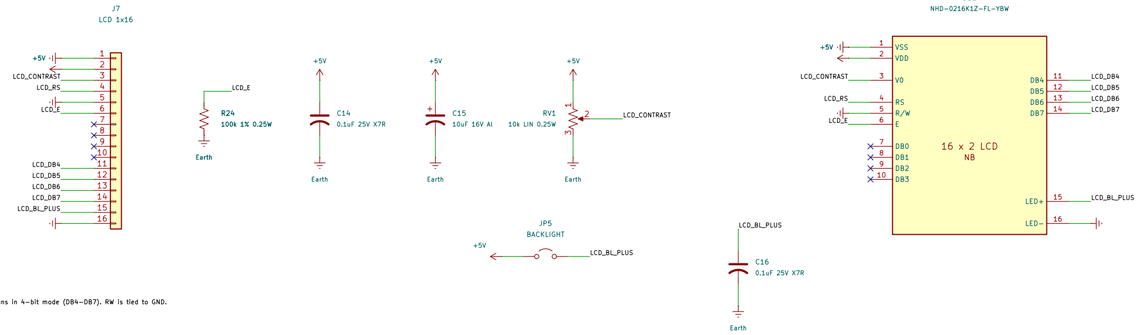


F1 is a 5x20mm fuse in a holder so it can be swapped.

Input is 9–12V DC, center positive.
U2 gets a heat sink (HSE14-384225P).

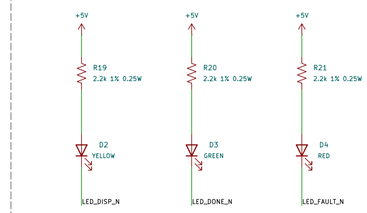


16x2 LCD

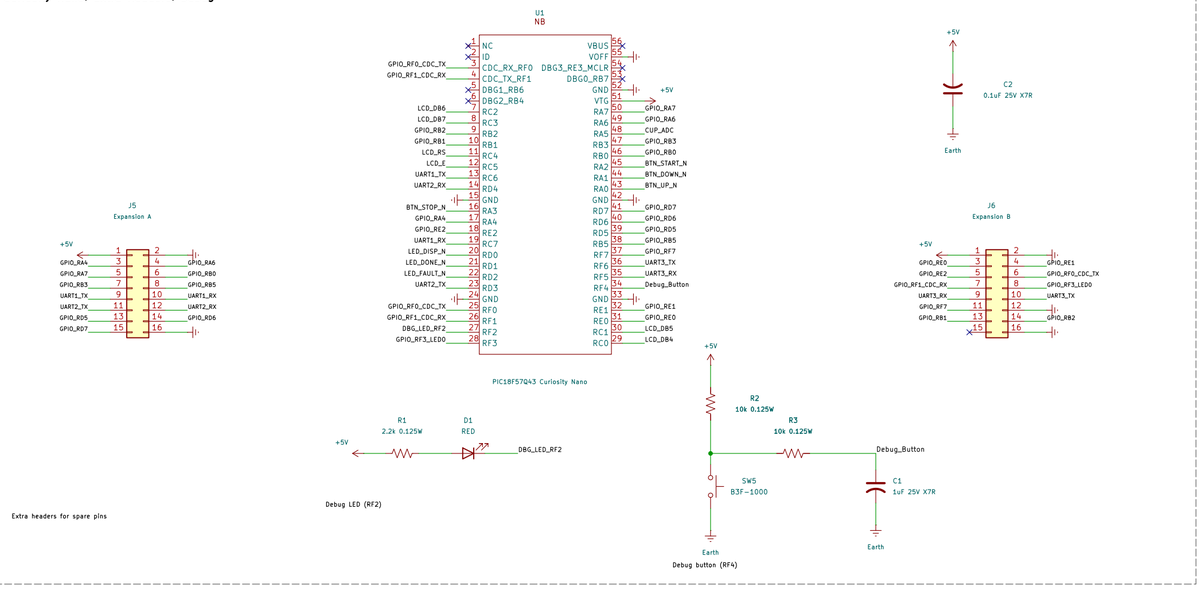


LCD runs in 4-bit mode (DB4–DB7). RW is tied to GND.

Status LEDs

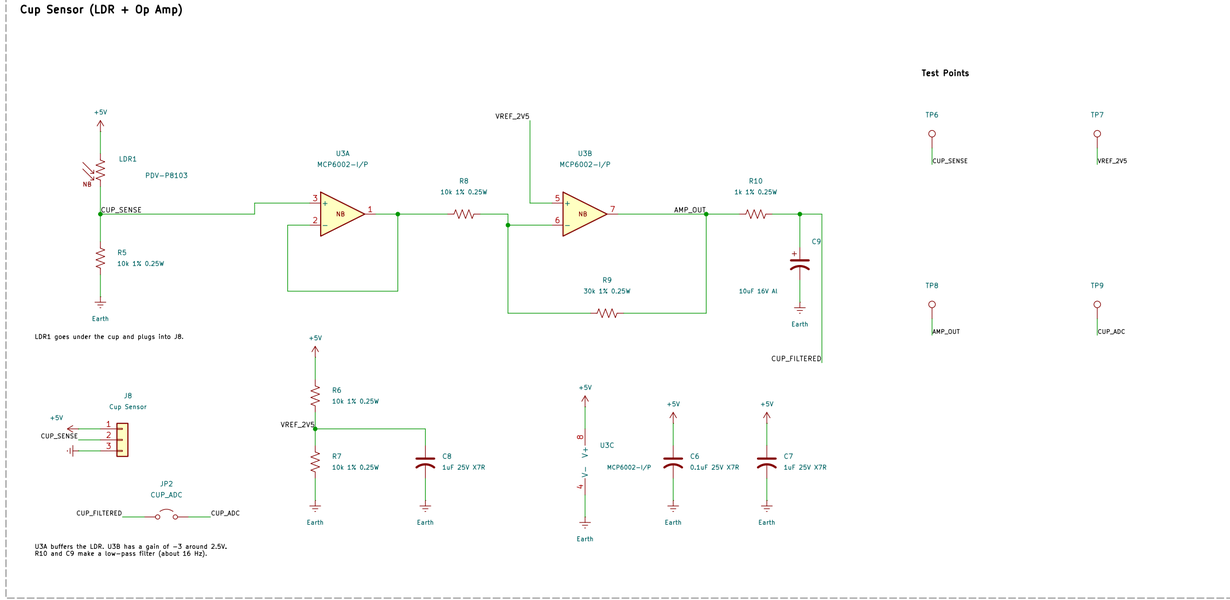


Curiosity Nano, Extra Headers, Debug



Extra headers for spare pins

Cup Sensor (LDR + Op Amp)

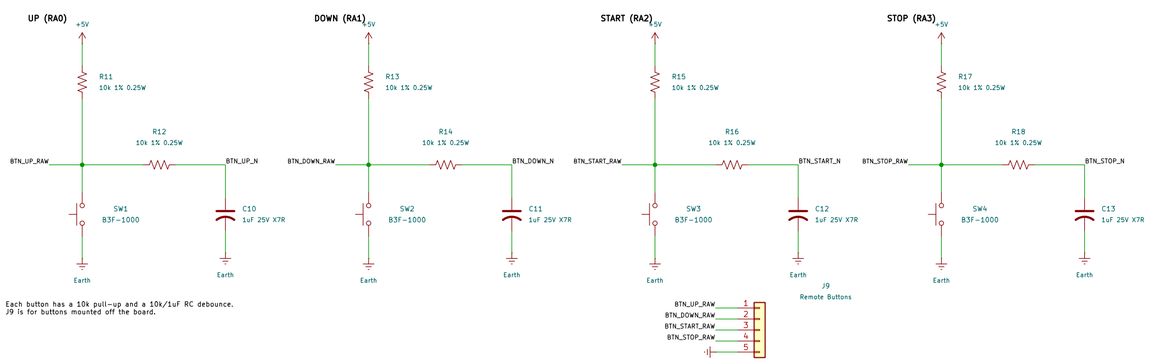


LDR1 goes under the cup and plugs into J8.

U3A buffers the LDR. U3B has a gain of −3 around 2.5V.
R10 and C9 make a low-pass filter (about 16 Hz).

Test Points

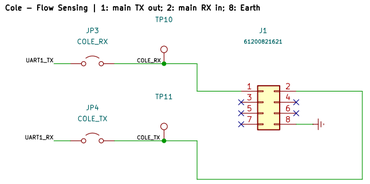
Buttons



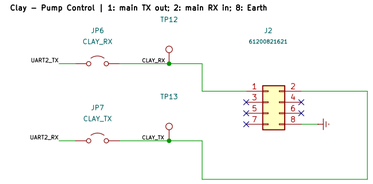
Each button has a 10k pull-up and a 10k/1uF RC debounce.
J9 is for buttons mounted off the board.

Ribbon Cables to Teammates (UART)

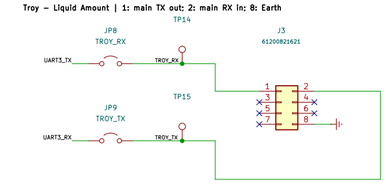
Cole — Flow Sensing | 1: main TX out; 2: main RX in; 8: Earth



Clay — Pump Control | 1: main TX out; 2: main RX in; 8: Earth



Troy — Liquid Amount | 1: main TX out; 2: main RX in; 8: Earth



Nikita Bangiyev — EGR 304 Team 201

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File: EGR304_Main_Control_Nikita.kicad_sch

Title: Fill-A-Bot Main Control Board

Size: A2	Date: 2026-10-19	Rev: 1
KiCad E.D.A. 10.0.6		Id: 1/1